

# CCE 1 Literature Survey

**Team code:** TY9 8\_A

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**Team Member 3:** Soham Phulare, 49, 7021985725

**Tentative Title:** Holistic Cryptocurrency Financial Analysis Using AI/ML Techniques

**Domain:** Cryptocurrency financial analysis using AIML

**Sub Domain:** AI/ML-Driven Cryptocurrency Financial Forecasting

**Objective Description:** To analyze and compare deep learning and sentiment-driven models for cryptocurrency price and volatility prediction, aiming to improve accuracy, reliability, and risk management in highly volatile markets.

## Team Member 1: Vansh Lilani

### PICO 1

- **Paper Title:** Probabilistic deep learning and transfer learning for robust cryptocurrency price prediction
- **Authors:** Amin Golnari, Mohammad Hossein Komeili, Zahra Azizi
  - **Problem Statement:** Cryptocurrency price forecasting is difficult due to extreme volatility and uncertainty, making deterministic predictions unreliable.
  - **Intervention:** A probabilistic GRU-based deep learning model enhanced with Bayesian neural networks and transfer learning.
  - **Comparison:** Compared against GRU, LSTM, probabilistic LSTM, and conventional forecasting methods.
  - **Outcome:** The proposed probabilistic approach outperformed alternatives, offering improved accuracy and robustness across multiple cryptocurrencies

### PICO 2

- **Paper Title:** Performance evaluation of deep learning and boosted trees for cryptocurrency closing price prediction
- **Authors:** Azeez A. Oyedele, Anuoluwapo O. Ajayi, Lukumon O. Oyedele, Sururah A. Bello, Kudirat O. Jimoh
  - **Problem Statement:** Cryptocurrency markets are highly volatile with complex datasets, limiting the effectiveness of traditional ML models for accurate price forecasting.
  - **Intervention:** Deep learning models (CNN, Deep Forward Neural Networks, GRU) optimized with genetic algorithms.
  - **Comparison:** Benchmarked against boosted tree models (GBM, Adaboost, XGB) across six cryptocurrency datasets from multiple sources.

- **Outcome:** CNN achieved the lowest error (MAPE 0.08) and highest variance score (0.96), proving more reliable with limited data and showing stronger generalization
- applicability by accounting for cross-coin dependencies.

### PICO 3

- **Paper Title:** Deep Learning-Based Cryptocurrency Price Prediction Scheme With Inter-Dependent Relations
- **Authors:** Sudeep Tanwar, Nisarg P. Patel, Smit N. Patel, Jil R. Patel, Gulshan Sharma, Innocent E. Davidson
  - **Problem Statement:** Most cryptocurrency forecasting models fail to capture interdependencies between smaller coins and their parent currencies, reducing real-time prediction accuracy.
  - **Intervention:** A hybrid deep learning model combining GRU and LSTM, incorporating parent coin direction (e.g., Bitcoin) for dependent cryptocurrencies (Litecoin and Zcash).
  - **Comparison:** Evaluated against standard standalone LSTM and GRU models.
  - **Outcome:** The hybrid model achieved higher forecasting accuracy, proving effective in real-time scenarios and supporting better decision-making for investors.

### PICO 4

- **Paper Title:** A Novel Cryptocurrency Price Prediction Model Using GRU, LSTM and bi-LSTM Machine Learning Algorithms
- **Authors of paper:** Hamayel, M.J., & Owda
- **Paper Description:**

This study applies deep learning methods (GRU, LSTM, bi-LSTM) to predict the prices of Bitcoin (BTC), Ethereum (ETH), and Litecoin (LTC) using historical price data. The models are evaluated using MAPE and RMSE, with GRU emerging as the most effective.

  - **Problem Statement:** Cryptocurrency prices are volatile and dynamic, making accurate forecasting difficult for investors and traders.
  - **Intervention:** Application of recurrent neural network architectures (GRU, LSTM, bi-LSTM) for price forecasting.
  - **Comparison:** Evaluated against each other (GRU vs. LSTM vs. bi-LSTM) on BTC, ETH, and LTC.
  - **Outcome:** GRU outperformed other models with the lowest MAPE, providing the most accurate predictions close to actual prices.

## PICO 5

- **Paper Title:** Probabilistic Deep Learning and Transfer Learning for Robust Cryptocurrency Price Prediction
- **Authors of paper:** Golnari, A., Komeili, M.H., & Azizi
- **Paper Description:**

This research proposes a probabilistic GRU (P-GRU) model enhanced with Bayesian neural networks and transfer learning for robust cryptocurrency forecasting. The approach generates probability distributions instead of point predictions, allowing better handling of uncertainty in volatile markets.

- **Problem Statement:** Cryptocurrency prices (especially Bitcoin) are highly volatile and uncertain, and traditional models struggle to incorporate prediction uncertainty.
- **Intervention:** Probabilistic GRU (P-GRU) with Bayesian inference and transfer learning across cryptocurrencies.
- **Comparison:** Benchmarked against GRU, LSTM, probabilistic LSTM, and other traditional deep learning models.
- **Outcome:** Achieved superior accuracy, robustness, and generalization to multiple cryptocurrencies (BTC, ETH, LTC, ADA, TRX, DOT, XLM) with improved MAPE,  $R^2$ , and explained variance scores.

## Team Member 2: Varadraj Mokashi

## PICO 1

- **Paper Title:** A novel hybrid cryptocurrency forecasting model integrating sentiment analysis and advanced neural networks
- **Authors of paper:** Adrian Vieitez, Matilde Santos, Rodrigo Naranjo
- **Paper Description:**
  - **Problem Statement:** Cryptocurrency prices are highly volatile and influenced by market sentiment, making traditional prediction methods unreliable.
  - **Intervention:** Hybrid forecasting model integrating sentiment analysis and advanced neural networks.
  - **Comparison:** Compared against standalone ML/DL models and conventional forecasting techniques.
  - **Outcome:** Achieved higher predictive accuracy and robustness, showing that sentiment-enriched hybrid models improve forecasting reliability.

## PICO 2

- **Paper Title:** Explainable deep reinforcement learning for financial trading strategies
- **Authors of paper:** Raúl Gomez-Martínez, Mara Luisa Medrano-Garcia
- **Paper Description:**
  - **Problem Statement:** Financial markets demand transparency in AI models; black-box DRL reduces trust in automated strategies.
  - **Intervention:** Explainable DRL framework applied to trading strategy development.
  - **Comparison:** Benchmarked against traditional DRL and non-explainable AI trading systems.
  - **Outcome:** Delivered competitive profitability while providing interpretability, making AI strategies more trustworthy for practitioners.

## PICO 3

- **Paper Title:** An Improved Machine Learning-Driven Framework for Cryptocurrencies Price Prediction With Sentimental Cautioning
- **Authors of paper:** Muhammad Zubair, Jaffar Ali, Musaed Alhussein, Shoaib Hassan, Khursheed Aurangzeb, Muhammad Umair
- **Paper Description:**
  - **Problem Statement:** Cryptocurrency forecasting suffers from errors due to volatility and neglect of sentiment-driven market dynamics.
  - **Intervention:** Multi-domain ML framework using Bi-LSTM + GRU for time-series forecasting and BERT + VADER for sentiment cautioning.
  - **Comparison:** Outperforms existing technical-only and sentiment-only models.
  - **Outcome:** Significantly lower RMSE values across BTC, ETH, and DOGE, plus sentiment-based alerts that improve investor risk management.

## PICO 4

- **Paper Title:** OPTICALS: A Novel Framework for Optimizing Predictive Trading Indicators in Cryptocurrency Using Advanced Learning Simulations
- **Authors of paper:** Hasib Shamshad, Fasee Ullah, Syed Adeel Ali Shah, Muhammad Faheem, Beena Shamshad
- **Paper Description:**
  - **Problem Statement:** Cryptocurrency markets are volatile; traders lack transparent, actionable forecasting tools with model interpretability.
  - **Intervention:** OPTICALS framework integrating ML (XGBoost, LightGBM) and DL (LSTM, Bi-LSTM, GRU) with hyperparameter optimization and look-back window parameters.
  - **Comparison:** Benchmarks gradient boosting models vs. recurrent neural networks under optimized training.
  - **Outcome:** GRU proved superior for day trading; XGBoost worked best for swing trading, offering practical tools for different investor strategies.

## PICO 5

- **Paper Title:** Cryptocurrency price prediction using machine learning and sentiment analysis
- **Authors of paper:** Susanna Levantesi , Gabriella Piscopo, Alba Roviello
- **Paper Description:**
  - **Problem Statement:** Crypto price forecasts using single-domain factors fail due to multi-factor market influences.
  - **Intervention:** BLGBV (Bi-LSTM + GRU + BERT + VADER) hybrid model with sentiment analyzer and alert mechanism.
  - **Comparison:** Evaluated against prior uni-domain ML/DL models that showed higher prediction errors.
  - **Outcome:** Produced highly precise predictions with RMSE of 0.0241% (BTC), 0.0645% (ETH), and 0.0978% (DOGE), outperforming benchmarks.

**Team Member 3: Soham Phulare**

## PICO 1

- **Paper Title:** Investor Sentiment and the Holiday Effect in the Cryptocurrency Market: Evidence from China
- **Authors of paper:** Pengcheng Zhang
- **Paper Description:**
  - **Problem Statement:** The cryptocurrency market in China shows abnormal return patterns around holidays, and investor sentiment may explain these anomalies.
  - **Intervention:** Construction of a sentiment indicator using textual analysis to capture mood effects.
  - **Comparison:** Market returns during holidays without sentiment adjustment (baseline holiday effect).
  - **Outcome:** Positive investor sentiment reduces the holiday effect on cryptocurrency returns, showing sentiment plays a mediating role.

## PICO 2

- **Paper Title:** Sentiment-Driven Cryptocurrency Forecasting: Analyzing LSTM, GRU, Bi-LSTM, and Temporal Attention Model (TAM)
- **Authors of paper:** Phumudzo Lloyd Seabe
- **Paper Description:**
  - **Problem Statement:** Bitcoin forecasting models often underperform because they neglect fine-grained sentiment from social media.
  - **Intervention:** Bi-LSTM model incorporating RoBERTa-based sentiment embeddings from Twitter.
  - **Comparison:** Other deep learning models (LSTM, GRU, TAM) using VADER sentiment features.
  - **Outcome:** Bi-LSTM with RoBERTa achieved the lowest MAPE (2.01%), proving the effectiveness of advanced NLP sentiment features.

## PICO 3

- **Paper Title:** Bitcoin Price Prediction Using Sentiment Analysis and Empirical Mode Decomposition
- **Authors of paper:** Serdar Arslan
- **Paper Description:**
  - **Problem Statement:** Bitcoin's price movements are complex and influenced by external sentiment factors, which are often excluded in time-series models.
  - **Intervention:** Parallel LSTM networks combining EMD-decomposed price data with Twitter sentiment signals.
  - **Comparison:** Price-only forecasting models without sentiment integration.
  - **Outcome:** Sentiment-augmented models significantly outperformed baselines, confirming the predictive power of social signals.

#### PICO 4

- **Paper Title:** Crypto Volatility Forecasting: Mounting a HAR, Sentiment, and Machine Learning Models
- **Authors of paper:** Alexander Brauneis
- **Paper Description:**
  - **Problem Statement:** Traditional models like HAR struggle to predict high volatility in cryptocurrencies due to their non-linear and sentiment-driven dynamics.
  - **Intervention:** Advanced ML/DL models (LSTM, LightGBM, CNN-BiLSTM, MLP, XGBoost) incorporating sentiment data.
  - **Comparison:** Standard HAR model without sentiment factors.
  - **Outcome:** ML/DL models outperformed HAR; incorporating sentiment further improved non-linear models' accuracy in volatility forecasting.

#### PICO 5

- **Paper Title:** Deep Learning for Bitcoin Price Direction Prediction: Models and Trading Strategies Empirically Compared
- **Authors of paper:** Alexander Brauneis
- **Paper Description:**
  - **Problem Statement:** Predicting Bitcoin's price direction is difficult; trading strategies based on weak models can be unprofitable.
  - **Intervention:** Deep learning architectures (CNN-LSTM, LSTNet, TCN) with feature selection techniques (Boruta, GA, LightGBM).
  - **Comparison:** ARIMA models and DL models without feature selection.
  - **Outcome:** CNN-LSTM with Boruta achieved 82.44% directional accuracy; backtested trading strategy yielded returns up to 6654% annually, far exceeding baselines.

Github Link: <https://github.com/Soham2805/Research-Practices>